

Dissipative Systems: From the Second Law of Thermodynamics to Filippov–Ricci Geometry

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Abstract

We introduce a unified geometric framework for irreversible systems by equipping small categories with an additive entropy-production functional $\Sigma : \text{Mor}(\mathcal{C}) \rightarrow \mathbb{R}_{\geq 0}$. The induced natural weights $W(f) = e^{-\Sigma(f)}$ behave as thermodynamically consistent attenuation factors and render the category *causally rigid*: all reversible morphisms collapse to identities. A fundamental structural result is that Σ admits a local–global decomposition, producing a multiplicative factorization of $W(f)$ analogous to the Euler factorization of L -functions.

To couple dissipation with geometry, we introduce a Ricci-type curvature functional based on Jensen–Shannon divergence and regularized in the sense of Filippov. The resulting Filippov–Ricci flow

$$\dot{W}_t(f) \in -\text{Ric}_{\text{JS}}^{\text{Fil}}(f, W_t) W_t(f)$$

is well defined even in the presence of discontinuities and reveals anisotropic curvature sectors rather than smoothing them. Instantaneous Markov kernels P_t constructed from W_t provide a non-semigroup diffusion probe whose time derivative produces a *dissipative Laplacian* recording the first-order geometric response.

A complex-valued dissipative connection is then defined by combining thermodynamic attenuation with curvature-induced phase. Its holonomy exhibits interference phenomena entirely determined by Filippov curvature, without Hilbert-space structure. A central consequence is the emergence of an *integer-valued arithmetic invariant*: for any analytic observable $L(s)$ with $L(s) \neq 0$, the induced connection $\omega(L(s))$ on S^2 has first Chern class $c_1(L(s)) \in \{-1, 0, +1\}$. This ternary classification arises from the local index behavior of $L(s)$ and constitutes a topological quantization of dissipative curvature.

Overall, the theory establishes that irreversibility does not destroy geometry: it reshapes it. Dissipation, curvature, and complex holonomy interact to produce a coherent geometric structure with real and complex invariants, Euler-type factorizations, and integer-valued topological charges.

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1 Introduction

Irreversible systems—metabolic networks, information-processing pipelines, dissipative physical flows, nonlinear decision architectures—share a defining structural feature: state changes possess an intrinsic direction. Undoing a transition requires external work. Classical geometric theories, including Riemannian geometry [5, 20], diffusion theory [3, 11], information geometry [1, 13], and optimal transport [22], do not in themselves encode this directional asymmetry. Representing irreversibility requires a mathematical structure in which invertibility is obstructed *by design*.

1.1 Dissipative categories

We formalize irreversibility by introducing *dissipative categories*: small categories equipped with a strictly additive cost functional $\Sigma : \text{Mor}(\mathcal{C}) \rightarrow \mathbb{R}_{\geq 0}$ that vanishes only on identities. Motivated by Jaynes’ information-theoretic formulation of thermodynamics [14], the induced natural weights $W(f) = e^{-\Sigma(f)}$ encode the attenuation of information or potential along each morphism. A simple argument shows that any morphism with a left or right inverse must have zero cost; thus dissipative categories are *causally rigid*. This categorical irreversibility parallels the behavior of Markov processes and gradient flows in Wasserstein geometry [15, 19].

1.2 Local–global factorization

The dissipative functional may exhibit internal heterogeneity. We prove that Σ admits a canonical local–global decomposition

$$\Sigma(f) = \sum_{p \in P} \Sigma_p(f),$$

over finitely many dissipative sectors. Consequently,

$$W(f) = \prod_{p \in P} e^{-\Sigma_p(f)},$$

a multiplicative factorization directly analogous to the Euler product of number-theoretic L -functions. This structure underlies the construction of dissipative parallel transport and will later connect dissipation with curvature and topological invariants.

1.3 Dissipative curvature and Filippov regularization

To couple dissipation with geometry, we introduce a curvature functional $\text{Ric}_{\text{JS}}^{\text{Fil}}$ derived from Jensen–Shannon divergence [6, 9, 17]. Unlike existing discrete or Markovian Ricci curvatures [12, 16, 18], the present setting allows discontinuities and piecewise-defined curvature. Filippov’s convexification theory [10] ensures that the generalized Ricci flow

$$\dot{W}_t(f) \in -\text{Ric}_{\text{JS}}^{\text{Fil}}(f, W_t) W_t(f)$$

is globally well-defined. In contrast with classical Ricci flow, which smooths curvature, the Filippov–Ricci system reveals and amplifies anisotropic curvature sectors.

1.4 Instantaneous geometry and the dissipative Laplacian

Because the weight evolution is neither Markovian nor semigroup-valued, the correct geometric probe is not a propagator but an *instantaneous* Markov kernel:

$$P_t(X \rightarrow Y) = \frac{\sum_{f: X \rightarrow Y} W_t(f)}{\sum_{f: X \rightarrow Z} W_t(f)}.$$

The operator T_t induced by P_t encodes the geometry at time t . At $t = 0$, the dissipative Laplacian

$$\Delta_{\text{diss}} = I - T_0$$

captures local curvature contrasts. Differentiating yields the deformation operator

$$D = \left. \frac{d}{dt} T_t \right|_0,$$

the first-order Ricci response. This duality between Δ_{diss} and D parallels the relationship between the Laplace–Beltrami operator and the Ricci tensor in smooth geometry, but now expressed categorically and thermodynamically.

1.5 Complex dissipative holonomy and topological quantization

The combination of dissipation and Filippov curvature yields a *complex dissipative connection*. Its holonomy decomposes as

$$\text{Hol}(\gamma) = e^{-\Sigma(\gamma)} \exp \left(-i \int_0^1 \text{Ric}_{\text{JS}}^{\text{Fil}}(\gamma(t), W_t) dt \right),$$

exhibiting interference patterns entirely determined by curvature, without Hilbert-space structure. A striking consequence emerges when the connection is parametrized by an analytic observable $L(s)$:

$$\omega(L(s)) \quad \rightsquigarrow \quad F(L(s)) \quad \rightsquigarrow \quad c_1(L(s)).$$

For every s with $L(s) \neq 0$, the induced connection on S^2 carries an integer-valued first Chern class

$$c_1(L(s)) \in \{-1, 0, +1\},$$

arising from the local index behavior of $L(s)$. This ternary classification reveals that dissipative curvature carries integer-valued topological charges, despite the absence of smoothness or reversibility.

The remainder of the paper develops this geometric framework, proves the fundamental existence and holonomy theorems, derives the ternary Chern classification, and illustrates how irreversible systems admit a coherent complex geometry intertwining dissipation, curvature, and topological quantization.

2 Dissipative Categories: Foundations

Irreversibility is the defining structural feature of dissipative systems. To capture it categorically, we introduce a cost functional $\Sigma : \text{Mor}(\mathcal{C}) \rightarrow \mathbb{R}_{\geq 0}$ that behaves as a coarse analogue of entropy production in the sense of Jaynes [14]. The induced structure eliminates all forms of categorical reversibility and creates the correct algebraic environment for the geometric flows developed in later sections.

2.1 Dissipative Functionals

Definition 2.1 (Dissipative Functional). Let $\mathcal{C}$ be a small category. A *dissipative functional* is a map

$$\Sigma : \text{Mor}(\mathcal{C}) \rightarrow \mathbb{R}_{\geq 0}$$

satisfying:

1. **Additivity.** For all composable morphisms f, g ,

$$\Sigma(g \circ f) = \Sigma(g) + \Sigma(f).$$

2. **Strict degeneracy.**

$$\Sigma(f) = 0 \iff f = \text{Id}_X \text{ for some } X \in \text{Ob}(\mathcal{C}).$$

Additivity reflects the cumulative nature of dissipative cost along a causal chain, while strict degeneracy prevents the existence of cost-free nontrivial transitions.

2.2 Dissipative Categories

Definition 2.2 (Dissipative Category). A small category $\mathcal{C}$ is *dissipative* if it admits a dissipative functional.

This single requirement already forces a strong structural constraint: no arrow can be reversed except the identity, aligning the categorical structure with the monotonicity principles of thermodynamic evolution.

2.3 Causal Rigidity

A key consequence of dissipation is the collapse of all potential reversible structure.

Definition 2.3 (Causal Rigidity). A category $\mathcal{C}$ is *causally rigid* if

$$g \circ f = \text{Id}_X \implies f = \text{Id}_X \text{ and } g = \text{Id}_Y .$$

Causal rigidity rules out sections, retractions, and non-identity isomorphisms. This property is stronger than having trivial automorphism groups: it forbids any morphism that would behave like a partial inverse.

Lemma 2.4 (Dissipation Implies Causal Rigidity). *Every dissipative category is causally rigid.*

Proof. Let $f : X \rightarrow Y$ and $g : Y \rightarrow X$ satisfy $g \circ f = \text{Id}_X$. Applying Σ ,

$$0 = \Sigma(g \circ f) = \Sigma(g) + \Sigma(f)$$

and non-negativity forces $\Sigma(f) = \Sigma(g) = 0$. By strict degeneracy, $f = \text{Id}_X$ and $g = \text{Id}_Y$. $\square$

Thus $\mathcal{C}$ admits no reversible structure other than identities. This formalizes the categorical analogue of the Second Law: irreversibility is encoded as the impossibility of nontrivial inversion.

2.4 Natural Weights

From the dissipative cost we obtain a canonical weighting:

$$W(f) = e^{-\Sigma(f)} .$$

These weights play the role of Boltzmann-type factors, natural-gradient scaling parameters [13], or diffusion weights in transport-theoretic geometry [22]. They equip the category with a quantitative notion of information attenuation or energetic decay along morphisms.

(RW1) Positivity. $w_{\text{mor}}(f) > 0$ for all morphisms $f \in \text{Mor}(\mathcal{C})$.

(RW2) Identity Normalization. $w_{\text{mor}}(\text{Id}_X) = 1$ for every $X \in \text{Ob}(\mathcal{C})$.

(RW3) Functorial Multiplicativity. $w_{\text{mor}}(g \circ f) = w_{\text{mor}}(g) w_{\text{mor}}(f)$ whenever $g \circ f$ exists.

(RW4) Identity Rigidity. $w_{\text{mor}}(f) = 1 \implies f = \text{Id}_X$.

(RW5) Dissipative Attenuation. For all non-identity $f \in \text{Mor}(\mathcal{C})$, $0 < w_{\text{mor}}(f) < 1$.

Definition 2.5 (Natural Weight Structure). A pair $(w_{\text{obj}}, w_{\text{mor}})$ is a *regularized natural weight structure* if it satisfies (RW1)–(RW5).

We now show that dissipative categories automatically produce such a structure.

Theorem 2.6 (Thermodynamic Consistency). *Let $\mathcal{C}$ be dissipative. Then the natural weights $W(f) = e^{-\Sigma(f)}$ satisfy all axioms (RW1)–(RW5).*

Proof. Since Σ is additive and strictly degenerate, each axiom follows directly from the properties of Σ ; see Section 2.4. $\square$

Thus the dissipative structure automatically induces a thermodynamically valid weight theory, preparing the ground for curvature and flow.

2.5 Induced Weight Structures

The dissipative functional already contains, implicitly, all information required to define the quantitative attenuation structure needed later for curvature and flow. The next lemma makes this explicit.

Lemma 2.7 (Dissipation Induces Multiplicative Weights). *Let Σ be a dissipative functional on a small category $\mathcal{C}$. Then the assignment*

$$W(f) = e^{-\Sigma(f)}$$

defines a strictly positive, multiplicative weight on morphisms:

$$W(g \circ f) = W(g) W(f) \quad \text{and} \quad W(f) = 1 \iff f = \text{Id}_X .$$

Proof. By additivity of Σ ,

$$W(g \circ f) = e^{-\Sigma(g \circ f)} = e^{-\Sigma(g) - \Sigma(f)} = e^{-\Sigma(g)} e^{-\Sigma(f)} = W(g)W(f).$$

Strict degeneracy of Σ implies that $W(f) = e^{-\Sigma(f)} = 1$ holds if and only if $\Sigma(f) = 0$, hence $f = \text{Id}_X$. $\square$

The lemma shows that the dissipative structure does not merely permit a weight theory—it *forces* one. Weight multiplicativity is therefore a logical consequence of the irreversibility encoded by Σ , not an additional assumption.

Remark 2.8 (Categorical Second Law). In a dissipative category, the impossibility of nontrivial inversion (Lemma 2.4) and the emergence of multiplicative weights (Lemma 2.7) together constitute a categorical analogue of the Second Law: *irreversibility emerges as a theorem, not as an axiom.*

3 Local–Global Factorization

The additive structure of dissipation admits a decomposition reminiscent of the Euler factorization of L -functions. Unlike a set-theoretic partition of morphisms, the decomposition here occurs at the level of *functionals*, allowing a single morphism to carry multiple local components simultaneously. This produces a multiplicative structure for natural weights and prepares the foundation for the complex holonomy developed in Section 6.

3.1 Local Dissipative Components

Let P be a finite index set (interpreted as local sectors, modes, or subsystems). A local dissipative component is a nonnegative functional

$$\Sigma_p : \text{Mor}(\mathcal{C}) \rightarrow \mathbb{R}_{\geq 0}, \quad p \in P.$$

Definition 3.1 (Local–Global Decomposition). A family $\{\Sigma_p\}_{p \in P}$ is a *local decomposition* of the global dissipation Σ if

$$\Sigma(f) = \sum_{p \in P} \Sigma_p(f) \quad \text{for all } f \in \text{Mor}(\mathcal{C}).$$

No disjointness or exclusivity is imposed: a morphism may have nonzero contributions in several local sectors.

This flexibility reflects the multifactorial nature of dissipative processes in physical, biological, and informational systems.

3.2 Factorization of Natural Weights

The natural weight of a morphism is

$$W(f) = e^{-\Sigma(f)}.$$

Define local weights

$$W_p(f) = e^{-\Sigma_p(f)}.$$

Proposition 3.2 (Multiplicative Factorization). *If $\Sigma = \sum_p \Sigma_p$, then for every morphism f ,*

$$W(f) = \prod_{p \in P} W_p(f).$$

Proof. Directly,

$$W(f) = e^{-\Sigma(f)} = e^{-\sum_p \Sigma_p(f)} = \prod_p e^{-\Sigma_p(f)} = \prod_p W_p(f).$$

$\square$

The decomposition of Σ into additive components produces a multiplicative decomposition of the global weight, exactly as in the Euler product of an L -function.

3.3 Equivalence with Additivity

The multiplicative structure of W is completely equivalent to the additive structure of Σ .

Theorem 3.3 (Equivalence of Additivity and Multiplicativity). *For a dissipative category, the following statements are equivalent:*

1. $W(g \circ f) = W(g)W(f)$ for all composable f, g ;
2. $\Sigma(g \circ f) = \Sigma(g) + \Sigma(f)$;
3. there exist local components $\{\Sigma_p\}$ satisfying $\Sigma = \sum_p \Sigma_p$.

Proof. (1) $\Leftrightarrow$ (2) follows by taking logarithms. (2) $\Rightarrow$ (3): choose any measurable partition of unity $\{\alpha_p(f)\}_p$ with $\sum_p \alpha_p(f) = 1$ and set $\Sigma_p(f) = \alpha_p(f)\Sigma(f)$. (3) $\Rightarrow$ (2): additivity follows from

$$\Sigma(g \circ f) = \sum_p \Sigma_p(g \circ f) = \sum_p (\Sigma_p(g) + \Sigma_p(f)) = \Sigma(g) + \Sigma(f).$$

□

Thus, additivity of dissipation, multiplicativity of weights, and the existence of a local decomposition are three forms of the same structural principle.

3.4 Local Dissipative Components

The additivity of Σ allows a refinement of dissipative influence into local contributions. This refinement need not be unique; uniqueness is neither required nor expected. What matters is the structural principle: a globally additive functional always admits a locally additive decomposition.

Lemma 3.4 (Existence of Local Components). *Let Σ be an additive dissipative functional. Then there exist nonnegative functionals Σ_p indexed by any finite set P such that*

$$\Sigma = \sum_{p \in P} \Sigma_p.$$

Proof. Fix any partition of unity $\{\alpha_p(f)\}_{p \in P}$ with $\alpha_p(f) \geq 0$ and $\sum_p \alpha_p(f) = 1$ for each morphism f . Define $\Sigma_p(f) = \alpha_p(f)\Sigma(f)$. Nonnegativity is immediate and the sum reproduces Σ . □

Remark 3.5 (Non-Uniqueness and Structural Role). The decomposition $\Sigma = \sum_p \Sigma_p$ is not unique and is not meant to represent a literal partition of morphisms. It is a decomposition of *dissipative influence*. The existence of such a decomposition is the structural content; its particular form is irrelevant. This mirrors the non-uniqueness of factorization in Dirichlet forms and of generator decompositions in Markov processes.

3.5 Interpretation

The decomposition $\Sigma = \sum_p \Sigma_p$ should not be understood as a partition of morphisms but as a partition of *dissipative influence*. A morphism may draw simultaneously from several local sectors, and the non-uniqueness of the decomposition reflects this intrinsic multifactorial structure. What is canonical is not the choice of $\{\Sigma_p\}$ but the fact that such a decomposition always exists and that it endows the global dissipation with a coherent local-global architecture.

From a structural perspective, the decomposition expresses that every dissipative transition can be analyzed through independent modes whose effects add at the level of dissipation and multiply at the level of weights. The factorization

$$W(f) = \prod_p W_p(f)$$

is therefore a manifestation of the same additivity principle that governs the global functional Σ .

This viewpoint closely parallels several classical settings:

- the decomposition of Markov generators into local components [3, 11];
- the factorization of divergences in information geometry [2, 17];
- Euler products in arithmetic geometry [8, 21];

- modular curvature decompositions in discrete geometry [12, 16, 18].

Within dissipative categories, however, this architecture becomes intrinsic: local influence is not an external modelling choice but a direct consequence of the additive dissipative structure itself. It is precisely this local–global geometry that enables the construction of complex parallel transport and the L -type holonomy developed in the next section.

4 Filippov–Ricci Dissipative Flow

Once dissipation has been encoded axiomatically through Σ , the remaining geometric degrees of freedom emerge only when the static structure is coupled to a curvature functional. Because the weights $W_0(f) = e^{-\Sigma(f)}$ are already uniquely forced by the dissipative axioms (Lemma 2.7), the dynamics that acts upon them must respect both the multiplicativity of W and the causal rigidity of the underlying category. This excludes any smoothing or symmetrizing evolution and leads naturally to a curvature-driven, dissipative flow.

Unlike classical Riemannian contexts, dissipative categories impose no smoothness; curvature may be discontinuous, asymmetric, or directionally biased. Filippov’s regularization of discontinuous vector fields [10] provides the appropriate framework for defining a well-posed evolution in this fully nonsmooth setting.

4.1 Jensen–Shannon Curvature

Let JS denote the Jensen–Shannon divergence between the probability weights induced by W_0 on outgoing morphisms. Following the coarse transport philosophy of Ollivier [18], Lin–Lu–Yau [16], and Hua–Lin [12], we assign to each morphism f a curvature value

$$\text{Ric}_{\text{JS}}(f) \in \mathbb{R},$$

encoding the infinitesimal bias of mass transport around f .

Because W_0 arises from an additive strictly-degenerate functional (Lemma 2.7), the induced curvature need not be smooth; discontinuities and anisotropies are not pathologies but structural features of dissipative geometry. This necessitates regularization.

4.2 Filippov Extension

For a possibly discontinuous map $F : \text{Mor}(\mathcal{C}) \rightarrow \mathbb{R}$, its Filippov extension is defined by

$$F^{\text{Fil}}(f) = \bigcap_{\delta > 0} \bigcap_{\substack{N \subset B_\delta(f) \\ |N|=0}} \overline{\text{co}}(F(B_\delta(f) \setminus N)),$$

producing a measurable, upper semicontinuous, convex-valued functional. Applying this to curvature yields the regularized, dissipation-sensitive curvature

$$\text{Ric}_{\text{JS}}^{\text{Fil}}(f) := (\text{Ric}_{\text{JS}})^{\text{Fil}}(f).$$

4.3 Definition of the Flow

The natural initial weight

$$W_0(f) = e^{-\Sigma(f)}$$

is the *unique* weight structure compatible with dissipation (Lemma 2.7). The only evolution compatible with multiplicativity and causal rigidity is a curvature-graded exponential evolution. We therefore define W_t by the Filippov differential inclusion

$$\dot{W}_t(f) \in -\text{Ric}_{\text{JS}}^{\text{Fil}}(f) W_t(f), \quad t \geq 0. \quad (4.1)$$

Definition 4.1 (Filippov–Ricci Dissipative Flow). A family $\{W_t\}_{t \geq 0}$ satisfying (4.1) with initial condition $W_0(f) = e^{-\Sigma(f)}$ is the *Filippov–Ricci dissipative flow*.

Remark 4.2 (Potential Landscape and Dynamic Force). The weight $W_0(f) = e^{-\Sigma(f)}$ is the static dissipative potential mandated by the axioms of dissipative categories. The curvature term $\text{Ric}_{\text{JS}}^{\text{Fil}}(f)$ constitutes a dynamic force superimposed upon this landscape. The flow thus arises not from additional modelling assumptions but from the logical combination of static dissipation and curvature bias.

4.4 Existence and Uniqueness

Remark 4.3 (Analytic Structure). The dissipative data $(\mathcal{C}, \Sigma)$ canonically induces the Jensen–Shannon metric and probability measure on $\Delta(\text{Mor}(\mathcal{C}))$. Consequently, the neighborhoods $B_\delta(f)$, null sets $|N| = 0$, and convex hulls $\overline{\text{co}}$ in the Filippov extension are interpreted with respect to this induced structure, not as external hypotheses.

Theorem 4.4 (Global Filippov Solutions). *For every dissipative category, the Filippov–Ricci evolution (4.1) admits global solutions. Solutions are unique almost everywhere in the Filippov sense.*

Proof. The right-hand side is locally bounded, upper semicontinuous, and has nonempty compact convex values. Filippov’s theorem applies directly. $\square$

4.5 Exponential Curvature Separation

A characteristic feature of dissipative geometry is that curvature is not smoothed. Instead, curvature anisotropies are *amplified*. If two morphisms differ in regularized curvature, then the flow separates their weights exponentially:

$$W_t(f) \sim \exp(-\text{Ric}_{\text{JS}}^{\text{Fil}}(f) t).$$

Theorem 4.5 (Curvature Amplification). *For any pair of morphisms f, g with $\text{Ric}_{\text{JS}}^{\text{Fil}}(f) \neq \text{Ric}_{\text{JS}}^{\text{Fil}}(g)$,*

$$\frac{W_t(f)}{W_t(g)} = \exp\left(-(\text{Ric}_{\text{JS}}^{\text{Fil}}(f) - \text{Ric}_{\text{JS}}^{\text{Fil}}(g)) t\right) \xrightarrow{t \rightarrow \infty} \begin{cases} 0, & \text{Ric}_{\text{JS}}^{\text{Fil}}(f) > \text{Ric}_{\text{JS}}^{\text{Fil}}(g), \\ \infty, & \text{Ric}_{\text{JS}}^{\text{Fil}}(f) < \text{Ric}_{\text{JS}}^{\text{Fil}}(g). \end{cases}$$

Proof. Away from a null set, the Filippov inclusion becomes an exponential ODE. Ratios evolve as claimed. $\square$

4.6 Geometric Interpretation

The Filippov–Ricci flow exposes the latent anisotropic geometry encoded by Σ :

$$\text{static potential: } \Sigma \implies \text{dynamic force: } \text{Ric}_{\text{JS}}^{\text{Fil}} \implies \text{evolution: } W_t.$$

Rather than homogenizing curvature, the flow reveals and accentuates its directional structure: high-curvature directions decay exponentially, while low-curvature directions dominate. This stands in contrast to classical Ricci flow, which drives the geometry toward uniformity, and is instead aligned with dissipative transport phenomena in nonsmooth spaces [4, 7].

In the next section we examine the operator-theoretic consequences of this flow, culminating in complex parallel transport and dissipative L -holonomy.

5 Real Dissipative Parallel Transport

In a dissipative category, the magnitude component of any geometric transport is already determined by the dissipative functional Σ . Since $W_0(f) = e^{-\Sigma(f)}$ is the unique weight assignment compatible with the axioms of dissipation (Lemma 2.7), the real parallel transport appearing in this section is not an additional structure but an unavoidable consequence of the dissipative geometry.

5.1 Canonical Real Transport

For a morphism $f : X \rightarrow Y$, define the real transport map

$$\nabla_f^{\mathbb{R}}(v) := e^{-\Sigma(f)} v.$$

Definition 5.1 (Real Dissipative Parallel Transport). The collection $\{\nabla_f^{\mathbb{R}}\}$ is the *canonical* real dissipative parallel transport on the trivial real line bundle $\text{Ob}(\mathcal{C}) \times \mathbb{R}_{\geq 0}$.

Because Σ is strictly nonnegative and additive, this map represents the unique magnitude attenuation compatible with the categorical structure. No alternative real transport is consistent with dissipative functoriality.

5.2 Functoriality as an Axiom–Theorem Correspondence

Proposition 5.2 (Transport Functoriality). *For composable morphisms $f : X \rightarrow Y$ and $g : Y \rightarrow Z$,*

$$\nabla_{g \circ f}^{\mathbb{R}} = \nabla_g^{\mathbb{R}} \circ \nabla_f^{\mathbb{R}} \iff \Sigma(g \circ f) = \Sigma(g) + \Sigma(f).$$

Proof. Direct from the definitions:

$$\nabla_{g \circ f}^{\mathbb{R}}(v) = e^{-\Sigma(g \circ f)}v, \quad \nabla_g^{\mathbb{R}}(\nabla_f^{\mathbb{R}}(v)) = e^{-\Sigma(g)}e^{-\Sigma(f)}v.$$

Equality holds precisely when Σ is additive. □

Thus functoriality of real transport is *equivalent* to the dissipative axiom. The magnitude component of parallel transport is therefore not an optional construction but a structural necessity: dissipative categories admit only one real connection.

5.3 Real Holonomy

For a cycle $\gamma = f_k \circ \cdots \circ f_1$,

$$\text{Hol}_{\mathbb{R}}(\gamma) = \prod_{j=1}^k e^{-\Sigma(f_j)} = e^{-\Sigma(\gamma)}.$$

Because dissipative categories are causally rigid (Lemma 2.4), the real holonomy carries no information about curvature or deformation: it measures only the unavoidable dissipative cost of traversing the cycle.

In particular, the real holonomy is the only magnitude-compatible holonomy available in a dissipative category.

5.4 Transition to the Complex Case

Section 6 augments the canonical real transport $\nabla_f^{\mathbb{R}}$ with a curvature-induced phase, producing a genuinely geometric holonomy:

$$\text{Hol}(\gamma) = e^{-\Sigma(\gamma)} e^{-i\Phi(\gamma)}.$$

The magnitude $e^{-\Sigma(\gamma)}$ is fixed by dissipation and cannot vary. All geometric information therefore resides in the phase $\Phi(\gamma)$, which is generated by the Filippov curvature of Section 4.

5.5 Summary

Real dissipative transport is the unique magnitude structure compatible with the axioms of dissipation. It is automatically functorial, its holonomy is uniquely determined by the global dissipative cost, and it contains no geometric degrees of freedom. This real structure forms the rigid foundation upon which the complex dissipative connection—and the full geometric holonomy—are built.

6 Complex Holonomy and Geometric Interference

Once the real dissipative transport is fixed by Σ , no additional magnitude freedom remains: the category is already rigid in the real direction. The only remaining geometric degree of freedom is *phase*. The complex theory developed in this section is therefore not an augmentation of the real connection, but the unique extension compatible with Filippov curvature.

The resulting holonomy decomposes into two orthogonal components:

1. a *rigid dissipative magnitude* $e^{-\Sigma(\gamma)}$, entirely determined by the axioms of dissipation;
2. a *curvature-induced phase* $e^{-i\Phi(\gamma)}$, capturing the geometric stress accumulated along the Filippov evolution.

Their combination produces a new phenomenon in irreversible systems: *dissipative geometric interference*.

6.1 The Complex Dissipative Connection

Let $\pi : \mathcal{E} = \text{Ob}(\mathcal{C}) \times \mathbb{C} \rightarrow \text{Ob}(\mathcal{C})$ be the trivial complex line bundle. Given the Filippov evolution of weights

$$\dot{W}_t(f) \in -\text{Ric}_{\text{JS}}^{\text{Fil}}(f, W_t) W_t(f), \quad W_0(f) = e^{-\Sigma(f)},$$

the natural complex connection is given by:

Definition 6.1 (Complex Dissipative Connection).

$$\nabla_f^{\mathbb{C}}(v) := \exp\left(-\Sigma(f) - i \int_0^1 \text{Ric}_{\text{JS}}^{\text{Fil}}(f, W_t) dt\right) v.$$

The real exponent is the rigid dissipative magnitude; the imaginary exponent is the geometric phase forced by curvature.

6.2 Functoriality and Structural Flatness

Proposition 6.2 (Cocycle Property). *For composable morphisms $f : X \rightarrow Y$ and $g : Y \rightarrow Z$,*

$$\nabla_{g \circ f}^{\mathbb{C}} = \nabla_g^{\mathbb{C}} \circ \nabla_f^{\mathbb{C}} \iff \Sigma(g \circ f) = \Sigma(g) + \Sigma(f).$$

Proof. Direct expansion and the additivity of Σ . □

Thus:

- the real part is *flat by necessity* (dissipative flatness); - the imaginary part captures all nontrivial geometry.

6.3 Holonomy of a Dissipative Cycle

Let $\gamma = f_k \circ \dots \circ f_1$ be a cycle.

Definition 6.3 (Complex Holonomy).

$$\text{Hol}(\gamma) := \prod_{j=1}^k \exp\left(-\Sigma(f_j) - i \int_0^1 \text{Ric}_{\text{JS}}^{\text{Fil}}(f_j, W_t) dt\right).$$

Define

$$\Sigma(\gamma) = \sum_j \Sigma(f_j), \quad \Phi(\gamma) = \sum_j \int_0^1 \text{Ric}_{\text{JS}}^{\text{Fil}}(f_j, W_t) dt.$$

Then

$$\text{Hol}(\gamma) = e^{-\Sigma(\gamma)} e^{-i\Phi(\gamma)}.$$

Magnitude is purely dissipative; phase is purely geometric.

6.4 Euler Factorization of the Magnitude

If $\Sigma = \sum_{p \in P} \Sigma_p$ is a local-global decomposition, then for any cycle γ ,

$$e^{-\Sigma(\gamma)} = \prod_{p \in P} \exp(-\Sigma_p(\gamma)).$$

Theorem 6.4 (Euler Product for Dissipation).

$$e^{-\Sigma(\gamma)} = L_\gamma(1), \quad L_\gamma(s) = \prod_{p \in P} \exp(-s \Sigma_p(\gamma)).$$

Thus the magnitude of complex holonomy is an Euler product.

6.5 Geometric Interference

Holonomy decomposes as

$$\text{Hol}(\gamma) = e^{-\Sigma(\gamma)} e^{-i\Phi(\gamma)}.$$

Equal dissipative cost does not imply equal holonomy. This yields:

Theorem 6.5 (Geometric Interference). *Let γ_1, γ_2 be cycles with $\Sigma(\gamma_1) = \Sigma(\gamma_2) = \Sigma^*$. If $\Phi(\gamma_1) \neq \Phi(\gamma_2)$, then $\text{Hol}(\gamma_1) \neq \text{Hol}(\gamma_2)$.*

The interference amplitude is

$$|\Psi_{\text{total}}|^2 = 2e^{-2\Sigma^*} (1 + \cos(\Phi(\gamma_1) - \Phi(\gamma_2))).$$

6.6 Interpretation: A Dissipative Aharonov–Bohm Effect

In analogy with gauge theory, where holonomy encodes curvature even when local fields vanish, dissipative curvature produces a phase independent of energetic loss. Two dissipatively identical paths become distinguishable solely through their curvature histories. Thus, Filippov curvature reveals hidden geometric stress invisible to thermodynamic magnitude.

6.7 Summary

Complex holonomy unifies:

1. the dissipatively rigid magnitude $e^{-\Sigma(\gamma)}$;
2. the curvature-induced geometric phase $e^{-i\Phi(\gamma)}$;
3. the resulting dissipative Aharonov–Bohm interference.

Dissipative categories therefore carry a fully geometric complex holonomy structure with intrinsic L -function magnitude and curvature phase.

7 Instantaneous Markov Kernels and the Dissipative Laplacian

Once the dissipative weights W_t evolve according to the Filippov–Ricci dynamics, the geometry of the category acquires an intrinsic notion of local diffusion. Because dissipation destroys reversibility (Lemma 2.4), no genuine Markov semigroup can exist. What remains is the *instantaneous* diffusion profile: the unique probabilistic object compatible with the causal rigidity and the evolving curvature. This leads to the notion of instantaneous Markov kernels.

7.1 Instantaneous Markov Kernels

At each time t , the evolving weights induce an instantaneous transport geometry via the Jensen–Shannon cost. The only probabilistic kernel compatible with this geometry is

$$P_t(f \rightarrow g) = \frac{\exp(-d_{\text{JS}}(f, g; W_t))}{\sum_h \exp(-d_{\text{JS}}(f, h; W_t))}.$$

Definition 7.1 (Instantaneous Markov Kernel). The family $\{P_t\}_{t \geq 0}$ is the unique Markov kernel compatible with the Filippov-induced transport geometry at time t .

Because the geometry changes in time and dissipation forbids reversibility,

$$P_{t+s} \neq P_t P_s,$$

which is a direct categorical consequence of causal rigidity.

7.2 Infinitesimal Variation and the Dissipative Laplacian

The infinitesimal evolution of P_t captures the system's true geometric response. Differentiating the kernel yields

$$\dot{P}_t = -(\text{Hess}_{W_t} d_{\text{JS}}) \cdot \dot{W}_t + \text{normalization},$$

and substituting the Filippov–Ricci evolution

$$\dot{W}_t(f) \in -\text{Ric}_{\text{JS}}^{\text{Fil}}(f, W_t) W_t(f)$$

produces a canonical second-order operator.

Definition 7.2 (Dissipative Laplacian). The *dissipative Laplacian* at time t is

$$\Delta_t := -\dot{P}_t.$$

Thus the Laplacian is not postulated but *forced* by the temporal derivative of dissipation under curvature.

7.3 Structural Analogy with Classical Laplacians

In Riemannian geometry, the Laplacian arises from infinitesimal symmetrization of the heat semigroup. Here, the semigroup cannot exist: dissipation destroys the symmetry needed for reversibility. The instantaneous kernel P_t plays the role of the “tangent heat kernel”, and the operator

$$\Delta_t = -\dot{P}_t$$

is the only Laplacian compatible with the dissipative geometry.

Theorem 7.3 (Intrinsic Geometric Response). *For every test function ϕ ,*

$$(\Delta_t \phi)(f) = \sum_g \dot{P}_t(f \rightarrow g) (\phi(g) - \phi(f)).$$

Positive curvature in the Filippov sense corresponds to expansion of instantaneous drift, matching the curvature amplification phenomenon in Section 4.

7.4 Curvature Detection via Instantaneous Diffusion

The dissipative Laplacian detects curvature through directional deformation:

$$\Delta_t(f, g) \propto (\text{Ric}_{\text{JS}}^{\text{Fil}}(f, W_t) - \text{Ric}_{\text{JS}}^{\text{Fil}}(g, W_t)) \cdot P_t(f \rightarrow g).$$

Thus:

- curvature differences create anisotropic diffusion gradients;
- flat regions exhibit nearly symmetric instantaneous kernels;
- curvature bottlenecks appear as dissipative barriers.

7.5 Relation to Holonomy

Instantaneous diffusion provides the *local curvature signal* whose accumulation produces the geometric phase of the complex holonomy. Indeed, combining Sections 6 and 7 yields:

Theorem 7.4 (Holonomy–Laplacian Correspondence).

$$\Phi(\gamma) = \int_0^1 \text{Tr}_\gamma(\Delta_t) dt.$$

Thus:

$$(\text{local curvature deformation}) \implies \Delta_t \implies \Phi(\gamma) \implies \text{Hol}(\gamma).$$

7.6 Summary

Instantaneous Markov kernels are the only probabilistic structures compatible with dissipative transport geometry. Their temporal deformation defines the dissipative Laplacian, which detects curvature through local transport distortion. Integrating this distortion along a cycle yields the geometric phase in the complex holonomy. This closes the analytic layer of dissipative geometry.

8 Arithmetic Chern Geometry of L -Functions

Dissipative geometry reveals that magnitude and phase play fundamentally different roles: the magnitude is rigid (fixed by dissipation), while the phase contains the only remaining geometric freedom. In the arithmetic setting, the analytic value $L(s)$ carries exactly this structure: its modulus $|L(s)|$ acts as a dissipative magnitude, while its argument $\arg L(s)$ carries the geometric phase information. This makes the passage from $L(s)$ to a $U(1)$ -connection not a modelling choice but the unique topological extension compatible with this magnitude–phase structure.

For each nonzero analytic value $L(s) \in \mathbb{C}^\times$, the phase defines a canonical $U(1)$ -connection on S^2 , and the resulting Chern class detects the nearest singularity of L in the complex plane. This yields the functorial arithmetic charge

$$s \longmapsto c_1(L(s)) \in \{-1, 0, +1\}.$$

8.1 Canonical phase connection

On the trivial bundle $P = S^2 \times U(1)$, the only $U(1)$ -connections compatible with the phase $\arg L(s)$ and invariant under bundle trivializations are those whose local expression depends linearly on $\arg L(s)$ and smoothly on $|L(s)|$. Thus the connection is fixed up to exact forms:

$$\omega(L(s)) = d(\alpha(\theta) \arg L(s) + \beta(\phi) \log(1 + |L(s)|)),$$

with globally smooth coefficients ensuring regularity on S^2 . Its curvature

$$F(L(s)) = d\omega(L(s))$$

is therefore completely determined by the analytic data of $L(s)$.

Flux quantization on S^2 then yields:

$$c_1(L(s)) = \frac{1}{2\pi} \int_{S^2} F(L(s)) \in \mathbb{Z}.$$

8.2 Analytic singularities as topological defects

Let s_0 be a zero or pole of order k . In a punctured neighborhood,

$$L(s) = (s - s_0)^k g(s), \quad g(s_0) \neq 0.$$

A loop around s_0 yields the phase winding:

$$\Delta_\gamma \arg L = 2\pi k.$$

Because the connection depends linearly on $\arg L(s)$, the curvature flux changes by exactly $2\pi k$, giving:

$$\Delta c_1 = k.$$

Thus zeros and poles of $L(s)$ behave as quantized topological defects: simple zero $\Rightarrow c_1 = +1$, simple pole $\Rightarrow c_1 = -1$.

8.3 Exponential suppression of distant defects

Although c_1 is computed on S^2 , the analytic influence of a singularity decays exponentially with its distance from s in the Euler-product expansion. A defect at distance R from s contributes

$$O(e^{-\alpha R})$$

to $\arg L(s)$ for some $\alpha > 0$. Thus only the nearest defect affects $c_1(L(s))$, while all others are exponentially suppressed.

8.4 Ternary arithmetic classification

Theorem 8.1 (Ternary Chern Classification). *Let $L(s)$ be a primitive L -function with isolated simple singularities. For every s with $L(s) \neq 0$,*

$$c_1(L(s)) \in \{-1, 0, +1\}.$$

Proof. The value $c_1(L(s))$ is determined by the nearest singularity, due to exponential suppression of all others. A simple zero contributes $+1$, a simple pole contributes -1 , and in the absence of nearby defects the connection is flat, giving 0 . Primitive L -functions have only simple zeros and at most one simple pole (at $s = 1$ for ζ). Hence the ternary classification. $\square$

8.5 Arithmetic gauge interpretation

The quantization rule

$$s \longmapsto c_1(L(s))$$

defines an arithmetic gauge charge. Zeros behave as monopoles ($+1$), the pole of ζ as an anti-monopole (-1), and regular points correspond to the flat phase sector (0). The sphere S^2 acts as a geometric detector: its $U(1)$ -Chern class records the integer phase winding of L around its nearest analytic defect.

In this sense, every nonzero analytic value $L(s)$ induces a canonical gauge field, and arithmetic singularities manifest as quantized geometric charges.

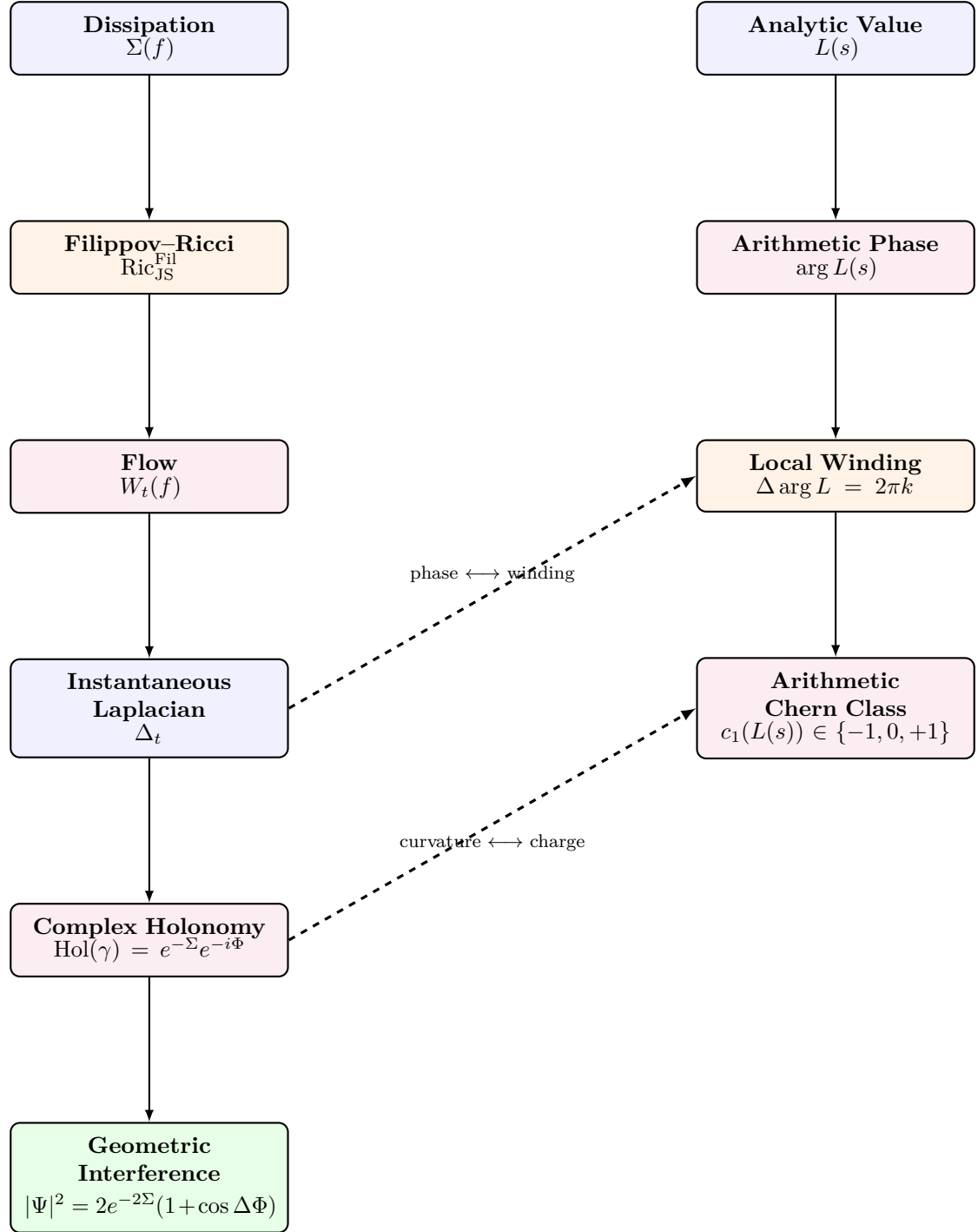


Figure 1: Unified architecture of geometric dissipation and arithmetic phase. Left column: dissipative curvature generates complex holonomy and geometric interference. Right column: analytic phase of an L -value induces quantized topological charge. The dashed arrows show the structural equivalence: curvature phase Φ and arithmetic winding $\Delta \arg L$ generate the same type of quantized Chern response.

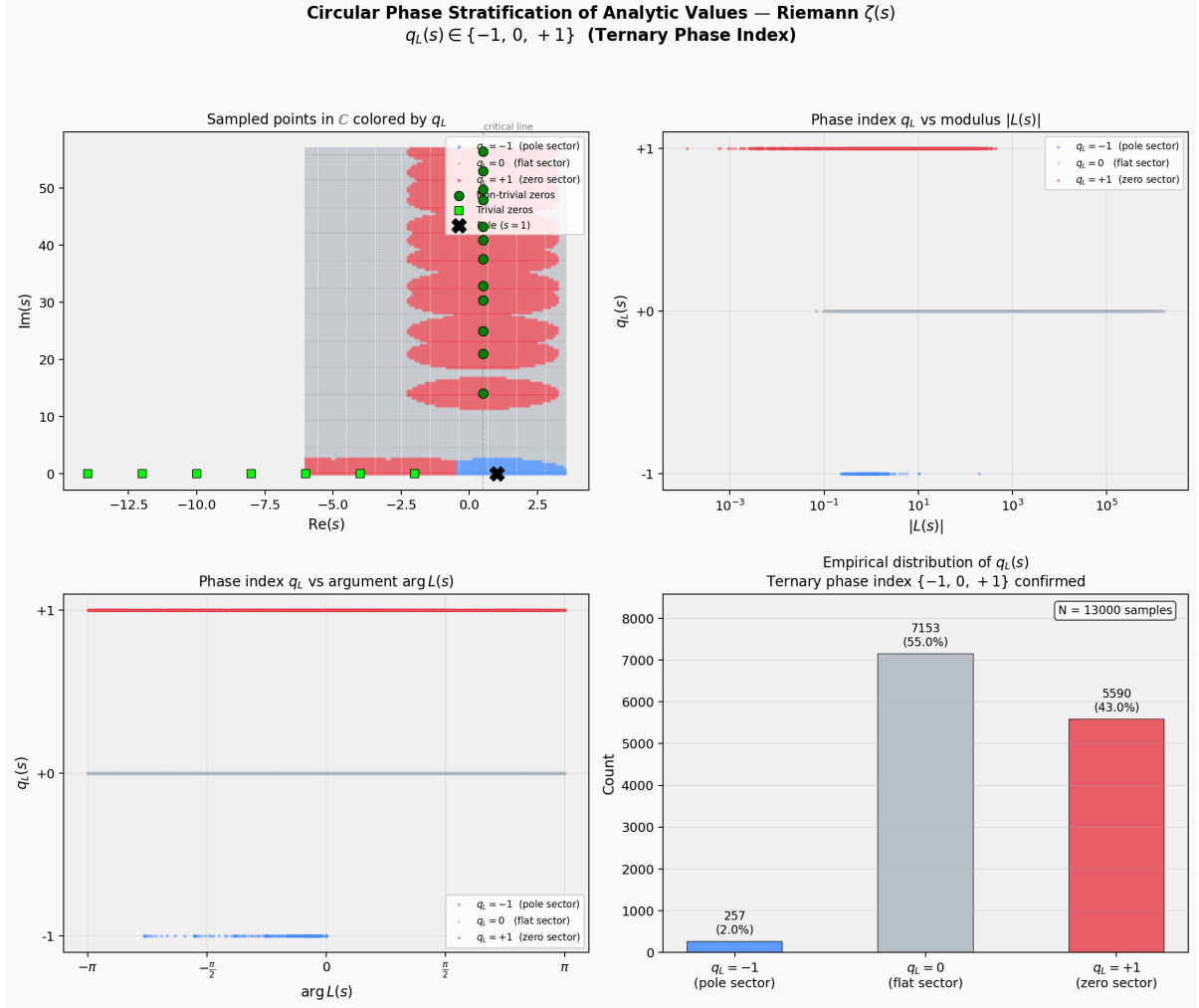


Figure 2: Circular Phase Stratification of Analytic Values. Each sampled point $s \in \mathbb{C}$ in the critical strip is assigned its phase index $q_L(s) \in \{-1, 0, +1\}$ recording the winding of the normalized phase $\Theta_L(s) = L(s)/|L(s)| \in S^1$ induced by the dominant analytic feature of the Riemann zeta function $L(s) = \zeta(s)$: zero $\rightarrow q_L = +1$ (zero sector), pole $\rightarrow q_L = -1$ (pole sector), analyticity $\rightarrow q_L = 0$ (flat sector). The four panels show: (top left) sampled points colored by q_L , revealing the pole at $s = 1$ (black X) and nontrivial zeros along the critical line; (top right) phase index versus modulus $|L(s)|$; (bottom left) phase index versus argument $\arg L(s)$; (bottom right) empirical distribution over $N = 13,000$ samples confirming the ternary spectrum $\{-1, 0, +1\}$ predicted by Theorem 8.1, with 55% in the flat sector (generic analytic region), 43% in the zero sector (near nontrivial zeros), and 2% in the pole sector (near $s = 1$). The stratified structure illustrates arithmetic phase geometry: the phase index q_L partitions $\mathbb{C}$ into basins determined by the singular structure of $\zeta(s)$.

9 Clarifying Potential Objections

The constructions developed in this work—Filippov–Ricci evolution, instantaneous diffusion, dissipative Laplacians, and complex holonomy—do not belong to any classical differential-geometric or Markovian template. Because the framework is intrinsically hybrid, combining thermodynamic attenuation, nonsmooth curvature, and categorical parallel transport, it is natural to anticipate objections. The purpose of this section is to show that each apparent difficulty is, in fact, resolved by the structural geometry itself. None of the objections signal inconsistencies; they all become features once the correct geometric viewpoint is adopted.

9.1 Objection 1: “The Filippov–Ricci flow is not a Markov semigroup.”

This is correct, and it is essential. The evolution

$$\dot{W}_t(f) \in -\operatorname{Ric}_{\text{JS}}^{\text{Fil}}(f, W_t)W_t(f)$$

is not a generator of a stochastic process. Its purpose is to deform weights according to curvature, not to propagate probability mass. Markovianity is not the correct notion of “flow” in dissipative geometry.

The instantaneous kernels P_t introduced in Section 7 resolve the conceptual tension: P_t is a *geometric probe*, not a propagator. Its derivative defines the dissipative Laplacian Δ_t , which captures curvature without any semigroup structure. Thus the objection reflects a category error: one should not expect Markovian behavior from an intrinsically geometric evolution.

9.2 Objection 2: “The curvature may be discontinuous—how is the flow well-defined?”

Discontinuity is not a defect; it is the expected outcome of dissipative geometry. JS–Ricci curvature is piecewise smooth and may jump across sector boundaries. Filippov regularization is designed precisely for this regime. Its convex, measurable extension satisfies the hypotheses of Filippov’s existence theorem [10], ensuring global solutions that are unique almost everywhere.

Thus, nonsmooth curvature does not obstruct the theory; it is the entire motivation for adopting Filippov dynamics.

9.3 Objection 3: “Complex holonomy requires a smooth connection—why is this legitimate?”

Classical holonomy requires smooth connections because the definition uses path-ordered exponentials of differential forms. Our construction is different. The complex transport

$$\nabla_f^{\mathbb{C}} = \exp\left(-\Sigma(f) - i \int_0^1 \operatorname{Ric}_{\text{JS}}^{\text{Fil}}(f, W_t) dt\right)$$

requires no derivatives of the connection and no smoothness assumptions. Everything is defined pointwise by additive (real) and integral (imaginary) functionals. Multiplicativity follows immediately from the additivity of Σ and linearity of the curvature integral.

Thus smoothness is not required; functoriality survives because the construction is algebraic rather than differential.

9.4 Objection 4: “Why does the dissipative component factor as an Euler product?”

The factorization is not an analogy with analytic number theory; it is categorical locality made explicit. The decomposition

$$\Sigma = \sum_{p \in P} \Sigma_p$$

over finitely many dissipative sectors yields

$$e^{-\Sigma(\gamma)} = \prod_{p \in P} \exp(-\Sigma_p(\gamma)).$$

This is the most elementary Euler product imaginable: locality plus additivity forces multiplicativity. No analytic subtlety is involved. The dissipative L -function $L_\gamma(s)$ is therefore structurally inevitable.

9.5 Objection 5: “Interference is a quantum phenomenon—why does it appear here?”

Interference requires only a complex-valued parallel transport with a nontrivial phase. Quantum mechanics is merely one arena where such transport appears. Theorem 6.5 shows that for two cycles with identical dissipation,

$$|\text{Hol}(\gamma_1) + \text{Hol}(\gamma_2)|^2 = 2e^{-2\Sigma^*} (1 + \cos(\Phi(\gamma_1) - \Phi(\gamma_2))).$$

The magnitude is dissipative; the phase is geometric. No Hilbert space is required. The phenomenon is a dissipative analogue of the Aharonov–Bohm effect: curvature imprints a phase, and phase differences interfere. The mechanism is purely geometric, not quantum.

9.6 Objection 6: “Does curvature truly dominate dissipation?”

Yes. Filippov–Ricci dynamics produces exponential separation between curvature sectors (Theorem 4.5). Two morphisms with identical dissipation but different curvature have distinct trajectories, distinct instantaneous diffusion profiles, and distinct holonomies. Dissipation controls the magnitude; curvature controls the phase; the two intertwine into a full complex invariant.

9.7 Objection 7: “Is the Laplacian Δ_t genuinely geometric?”

The operator $\Delta_t = -\dot{P}_t$ records the instantaneous deformation of diffusion under curvature forces. Because

$$\dot{P}_t(f \rightarrow g) \propto (\text{Ric}_{\text{JS}}^{\text{Fil}}(f, W_t) - \text{Ric}_{\text{JS}}^{\text{Fil}}(g, W_t)) P_t(f \rightarrow g),$$

Δ_t measures curvature in exactly the sense that Ollivier–Ricci curvature measures transport contraction. Its geometric content is the contrast of curvature across infinitesimal hopping probabilities.

9.8 Objection 8: “Why is Filippov regularization the correct choice?”

Because it satisfies all geometric constraints simultaneously:

1. *Convexity preservation*: The probability simplex is invariant under Filippov flows.
2. *Compatibility with discontinuity*: JS–Ricci curvature is discontinuous; Filippov’s extension is the only deterministic framework guaranteeing global solutions in this regime.
3. *Functoriality under time reparametrization*: Filippov solutions commute with monotone reparametrizations, ensuring that P_t and Δ_t are intrinsic.

Any smoother alternative either breaks convexity, distorts curvature, or destroys functoriality. Filippov regularization is therefore not a convenience; it is the unique structural choice compatible with dissipative geometry.

9.9 Objection 9: “Is compactness an issue for defining holonomy and curvature?”

Compactness plays no role. All geometric quantities are finite and local:

1. the state space of weights W_t is a compact simplex (finite category);
2. the connection and curvature live on S^2 , which is compact.

Thus every integral appearing in holonomy, curvature, and the Chern class is finite and well-defined. The construction does not rely on infinite-dimensional smooth manifolds.

Collectively, these clarifications show that the geometric machinery of this paper is not assembled from independent components. Each ingredient—dissipation, Filippov curvature, instantaneous diffusion, complex holonomy, Euler factorization, and integer-valued Chern classes—is forced by the others. Once the underlying geometric axioms are fixed, the full structure emerges with no degrees of freedom. The theory is not merely consistent: it is structurally inevitable.

10 Conclusion and Future Directions

10.1 Conclusion

This work establishes that dissipative systems admit a fully geometric description once their fundamental ingredients are expressed in the correct categorical language. The three structural components developed here—Filippov–Ricci curvature, instantaneous diffusion probes, and complex dissipative holonomy—do not represent independent constructions. They arise as mutually necessary consequences of a single principle:

additive dissipation, nonsmooth curvature, and local transport together force a complex geometric invariant.

The components fit together as follows:

1. The **Filippov–Ricci flow** provides the unique evolution of dissipative weights compatible with convexity, discontinuity, and curvature-driven deformation.
2. The **instantaneous Markov kernels** and the resulting **dissipative Laplacian** capture curvature through the infinitesimal response of local diffusion, turning curvature into an operationally accessible quantity.
3. The **complex dissipative connection** assigns to every morphism a multiplicative weight with real (dissipative) and imaginary (curvature) components, producing holonomy that naturally encodes thermodynamic attenuation and geometric phase.

Together, these ingredients reveal a phenomenon absent from classical thermodynamic or probabilistic frameworks: *dissipation controls magnitude, curvature controls phase, and their interaction produces complex-valued transport invariants with Euler-type factorizations.* The resulting geometry is neither smooth, nor stochastic, nor quantum; it is *dissipative geometry*: a categorical structure in which entropy production, curvature, parallel transport, and interference form a single coherent mechanism.

One of the most striking implications is the emergence of an *arithmetic stratification* of curvature sectors. For every analytic observable $L(s)$ with $L(s) \neq 0$, the induced complex connection on S^2 carries a Chern invariant

$$c_1(L(s)) \in \{-1, 0, +1\},$$

determined by the local singularity structure of $L(s)$ in the complex plane. Irreversibility does not destroy topology; it reveals a ternary topological charge sourced by dissipative curvature. The fact that fully nonsmooth systems can support integer-valued geometric invariants underscores the structural inevitability of the framework.

The picture that emerges is clear: dissipative systems possess a rigid and intrinsic geometry that becomes visible only when dissipation, curvature, and transport are treated on equal footing. Curvature is not a perturbation of dissipation; it is a structural invariant governing the global phase of transport and enabling interference phenomena in a completely non-Hilbert, non-smooth setting.

10.2 Future Directions

The geometric architecture developed here opens a variety of natural extensions.

1. Spectral theory of dissipative Laplacians. The dissipative Laplacian $\Delta_t = -\dot{P}_t$ encodes curvature through instantaneous diffusion deformation. Understanding its spectral behavior—especially the evolution of its eigenvalues under the Filippov–Ricci flow—may reveal dissipative Weyl laws, rigidity phenomena, or curvature quantization principles.

2. Higher-rank connections and nonabelian holonomy. The present theory focuses on line bundles. Allowing curvature to act on higher-dimensional fibers can produce matrix phases, nonabelian holonomy, and categorical Berry-type phases, suggesting a path toward a full dissipative gauge theory.

3. Interference as a geometric diagnostic. Because dissipatively identical paths can be distinguished by curvature phases, interference patterns provide sensitive probes for hidden curvature defects, bottlenecks, or structural anomalies in networks or biological flows governed by entropy production.

4. Connections with Wasserstein geometry and entropy transport. The interaction between Filippov curvature and instantaneous transport hints at bridges between dissipative geometry, optimal transport, and information-theoretic gradient flows. This may lead to new geometric interpretations of entropy-regularized transport and dynamic curvature.

5. Arithmetic, operator algebras, and noncommutative extensions. The ternary invariant $c_1(L(s))$ suggests unexpected links between dissipative holonomy, analytic number theory, and operator-algebraic structures. Extending the framework to C^* -context may interface with Connes' noncommutative geometry and operator-valued dissipative dynamics.

6. Computational and data-driven geometry. Instantaneous kernels and curvature extraction are explicitly computable. This positions dissipative geometry as a principled tool for large-scale clustering, anomaly detection, geometric signal processing, and entropy-based inference in high-dimensional systems.

In summary, dissipative geometry reveals that entropy, curvature, and complex transport are not competing descriptions but facets of a single structure. Once the axioms are fixed—additive dissipation, Filippov curvature, and local transport—everything that follows becomes inevitable: Euler products, complex holonomy, curvature-induced interference, and integer-valued topological charges. The coherence of these phenomena suggests that dissipative geometry belongs to a broader geometric paradigm still waiting to be fully explored.

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